

Load Interconnection System Impact Study Report

For Queue Project Q18-1051

Forman 100 kV
November 22, 2019

1. Introduction

This System Impact Study report documents the results of the analyses performed for interconnection request Q18-1051. The Interconnection Customer proposes a 128.5 MW large load (data center) facility interconnecting at the Forman 100 kV substation in the Illinois area. The request was received on August 30, 2018. The study evaluated system intact and single contingency conditions using the regional planning case, including 392 higher-queued active requests and their assigned network upgrades.

2. Summary of Results

The project was found to contribute (DFAX of 5% or greater) to 6 overloaded facilities. The total estimated cost of the network upgrades allocated to this project is \$12,470,000.

3. Network Impacts

3.1 Load Deliverability Analysis

Monitored Facility	Contingency	Loading (%)	Rating (MVA)	DFAX (%)
LAWRENCE HILL to CANADYS STEAM 100 KV	CANADYS STEAM to LAWRENCE HILL 100 KV #1	136.5	147	13.4
CANADYS STEAM to LAWRENCE HILL SUBSTATION 100 KV Line 1	LAWRENCE HILL to CANADYS STEAM 100 KV Ckt 2	132.4	152	13.4
PEACH GROVE D.P to FORMAN 100 KV Ckt 1	PEACH GROVE D.P to FORMAN 100 KV Line 2	123.0	151	41.2
CRESTWOOD to CANADYS STEAM 100 KV Circuit 1	CANADYS STEAM to CRESTWOOD 100 KV	111.3	146	7.5
CRESTWOOD to CANADYS STEAM 100 KV Line 2	CANADYS STEAM to CRESTWOOD 100 KV #1	111.3	146	7.5
PEACH GROVE D.P to FORMAN 100 KV	FORMAN to PEACH GROVE D.P 100 KV	102.7	180	41.2

3.2 Multiple Facility Contingency Analysis

No additional overloads were identified for multiple facility contingencies.

3.3 Short Circuit and Stability

Short circuit and stability analyses identified no additional network upgrades.

4. Required Network Upgrades and Cost Estimates

Facility	Required Upgrade	Cost Estimate	Allocation to Project
LAWRENCE HILL to CANADYS STEAM 100 KV	Rebuild line with higher-capacity conductor	\$7,870,000	54% (\$4,217,000)
CANADYS STEAM to LAWRENCE HILL 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$8,200,000	54% (\$4,394,000)

Facility	Required Upgrade	Cost Estimate	Allocation to Project
PEACH GROVE D.P to FORMAN 100 KV Ckt 1	Rebuild line with higher-capacity conductor	\$1,644,000	71% (\$1,170,000)
CRESTWOOD to CANADYS STEAM 100 KV Circuit 1	Rebuild line with higher-capacity conductor	\$3,070,000	38% (\$1,179,000)
CRESTWOOD to CANADYS STEAM 100 KV Line 2	Rebuild line with higher-capacity conductor	\$1,877,000	38% (\$721,000)
PEACH GROVE D.P to FOMAN 100 KV	Reconductor line	\$1,108,000	71% (\$789,000)
	Total allocated to project		\$12,470,000

This report is a simulated study produced by the gridconstraint research prototype on a public synthetic-but-realistic PJM-footprint case; it is not a PJM document.